RUSHIL BARVE

Computer Science & Computational Mathematics · University of Waterloo



About Me

CS & Computational Mathematics student seeking a Winter/Spring co-op. I ship full-stack projects, lead teams and events, and communicate clearly under pressure.

Contact

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LOCATION
Waterloo, Ontario

Skills

LANGUAGES
Python, JavaScript, TypeScript, GDScript, C, SQL, HTML, CSS
FRAMEWORKS
React, Vite, Tailwind CSS, FastAPI, Godot 4, Zustand
TOOLS
Git, GitHub Actions, SQLite, IndexedDB, Render

Achievements

- Gold Medal of Distinction, International Informatics Olympiad (2023-24); Bronze (2024-25)
- Gold Medal of Distinction, National Science Olympiad
- Six Best Delegate awards across 8+ national and international MUN conferences
- Basketball, CBSE Qatar Clusters: Gold (2025-26), Bronze (2023-24, 2024-25)

Certifications

- Python Essentials 2 · HTML Essentials · Computer Hardware Basics
- Accenture Nordics – Software Engineering Job Simulation
- Bloomberg – Client Service Job Simulation

EDUCATION

University of Waterloo — Expected 2031
Computer Science (Co-op) · Computational Mathematics

Delhi Public School – Modern Indian School, Doha — Graduated 2026
CBSE Senior Secondary, Computer Science

PROJECTS

CircuitGuard — Python · FastAPI · React · Three.js
Winner, Ampere – AI for Motor Safety · FormulaTech Hacks 2026
- Real-time Formula 1 safety system: analyses live driver telemetry to detect crashes and stopped cars on the racing line and recommends the FIA response (safety car, VSC, red flag) under the 2026 sporting regulations.
- Models per-sector spectator risk with a Bayesian Monte Carlo insurance-pricing model, so premiums can reflect each track section instead of a blanket rate.

Whisper Chain — Godot 4 · GDScript
Puzzle game built on function composition
- Puzzle game where each villager garbles a message (reverse, swap, double, drop a word); players route the message so the distortions cancel, discovering inverses and non-commutativity.
- Graph solver (DFS over simple paths) finds every correct route, sets par and powers live hints; used to design and validate all 10 levels.

MUN Chair Toolkit — TypeScript · React · Vite · Tailwind · Zustand · Dexie
Local-first web app for Model UN chairs · live on GitHub Pages
- Committee-management app for Model UN chairs: committee and agenda setup, bulk delegation entry, and roll call with Present, Present and Voting, and Absent states.
- Automated quorum, simple-majority and two-thirds calculations, so chairs no longer count votes by hand.

Source code available **on request** · more projects at rsk0110.github.io/s2rkportfolio

LEADERSHIP & EXPERIENCE

Director of MUN & Electoral and Democratic Club — May 2025 – Jun 2026
DPS-Modern Indian School, Doha
- Led and managed the school's MUN program: organized conferences, trained delegates, promoted diplomacy and public speaking.

Head of Marketing, TEDxYouth@DPSMIS — Sep – Dec 2025
- Led branding, social media and promotion for a TEDx-licensed youth event, working across curation, design, operations and technical teams.

Head of Operations, DPSDQ MUN — Jun – Oct 2025
- Ran logistics and delegate services for the 25th-anniversary edition, coordinating with the Secretariat.

Team Leader, Qatar Children's Science Congress — Dec 2024 – Mar 2025
- Led an original student research project from experimental design to data analysis and expert-panel presentation.

Co-Founder, HuAr (Humanities.Arts) — Feb 2024 – Mar 2025
- Co-founded a student organization guiding students on humanities and arts paths; organized conferences and speaker sessions.

Head of Outreach & Logistics, Q-SOMA — Jun 2025 – Jun 2026
Qatar Society of Medical Aspirants
- Led outreach and logistics and supported event management for the society.

MUN Trainer & Head Chair — May 2024 – May 2026
DPS-MIS · Catalyst MUN · Global MUN Network
- Trained delegates in research, procedure and resolution drafting; Head Chair of Historical Security Council (Catalyst MUN) and WHO (Global MUN Network).

WATERLOO TEAMS

WARG — Waterloo Aerial Robotics Group — 2026
Autonomy team · Embedded Flight Software team

UW Orbital · UW Game Dev — 2026
Student satellite design team · built beginner-level games

Member: Computer Science Club, UW Data Science Club